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## **A Novel Field Development Strategy: Case Study on the Effective Application of Smart Completions**

Djamal Kherroubi, Djelloul Hammadi, Yunus Berdiyev, Eva Sanchez, Sara Alhammadi, Mhd Maher Ahmed Alrefaai, and Amit Kumar, ADNOC Onshore, Abu Dhabi, United Arab Emirates

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### **Abstract**

The effective development of complex reservoirs requires completion strategies that enable both selective zonal production and controlled commingled production across multiple reservoirs. Smart completion technologies provide the capability to achieve this dual objective by integrating inflow control devices, zonal isolation, and real-time monitoring systems.

This case study presents the deployment of ADNOC Onshore's first smart completion for dual-reservoir production, designed to optimize selective inflow control while enabling commingled production. The installation incorporated Inflow Control Valves (ICVs), Permanent Downhole Gauges (PDHGs), and dual-sensor monitoring to achieve enhanced zonal isolation, dynamic production balancing, and improved well integrity. The solution facilitated real-time inflow/outflow control, reducing intervention requirements and enhancing sustained reservoir contribution from both zones.

To maximize the value of smart completions, a hybrid physics and data-driven workflow was implemented. This workflow combined proxy reservoir models, well performance modelling, and data driven surveillance tools to deliver continuous inflow monitoring, zonal allocation, and production optimization. The integration of physics-based analysis with data analytics enabled proactive identification of production anomalies, efficient water cut and GOR management, and reliable allocation of commingled volumes across the two reservoirs.

Results obtained demonstrated improved production performance, delayed water breakthrough, and higher well availability compared with conventional completions. Operational savings were realized through the elimination of rig-based workovers, reduced reliance on wireline logging for zonal contribution, and optimized reservoir management through automated surveillance and control.

This case study highlights a novel field development strategy that integrates smart completion hardware with advanced hybrid workflows. The demonstrated approach provides actionable best practices for operators seeking to enhance recovery efficiency, ensure reliable surveillance, and optimize dual-reservoir management in complex field environments.

## Introduction

Effective water and gas management is a key requirement for optimizing reservoir performance, extending well life, and improving hydrocarbon recovery. Among the range of subsurface technologies available, smart completions play a pivotal role by enabling selective inflow control, reducing unwanted water production, and providing the flexibility to commingle production from multiple reservoirs.

ADNOC Onshore has implemented several smart completion systems across its assets, including hydraulic Inflow Control Valve (ICV) completions. In the case study presented here, an ICV-based smart completion was deployed to address the operational challenges commonly encountered in dual-string wells, such as limited workover options and well integrity constraints. This deployment enabled production from two distinct reservoirs through a single completion string, eliminating the risk of long-string abandonment and adding approximately 300 BOPD of incremental production.

Beyond the hardware, the success of smart completions depends on effective surveillance and production allocation. To this end, an in-house hybrid physics and data-driven workflow was developed and implemented. This workflow integrates reservoir physics with advanced data analytics to ensure accurate real-time monitoring and zonal contribution tracking. Virtual flow meters (VFM) derived from surface and subsurface data including sandface measurements, ICV settings, and well completion parameters are used as base inputs. These VFMs are combined with data driven algorithm to predict phase behavior, allocate zonal production, and proactively identify reservoir or well condition changes.

Accurate zonal flow rate estimation is critical for both reservoir management and the evaluation of ICV performance. In this approach, zonal allocation is continuously updated in response to operational changes such as valve position adjustments or evolving well conditions. This hybrid workflow ensures seamless surveillance, improves decision-making, and supports proactive reservoir management. Benchmarking against well test data and production logging confirmed the robustness and reliability of the workflow.

This case study therefore illustrates not only the practical deployment of ICV-based smart completions in a dual-reservoir context, but also the integration of advanced hybrid workflows that enhance surveillance, optimize zonal allocation, and improve overall field development outcomes.

## Field Background

The studied well is a dual-string producer with a deviated wellbore, historically classified as an inactive asset due to significant operational failures in both strings. Major integrity challenges including SAP B leaks, string communication issues, debris blockage, and restricted mechanical access resulted in reduced operability and limited recovery potential. Workover difficulties associated with dual-string completions often force operators to either convert such wells to single-zone producers or abandon one of the producing intervals.

In this case, a detailed reservoir flow potential evaluation supported the decision to intervene and reactivate the well using the latest smart completion technologies, rather than permanently abandoning one of the zones. The objective was to restore zonal contribution, mitigate integrity concerns, and extend the productive life of the asset.

The proposed completion solution was designed to enable sustained production from both reservoirs within a single wellbore. Instead of limiting the well to a single-zone oil producer, a hydraulic Inflow Control Valve (ICV) smart completion was deployed in the lower reservoir to restore approximately 300-400 BOPD as part of the recovery plan. Due to the high water cut (~70%) in the lower zone and the absence of a gas lift tie-in, production sustainability was ensured by commingling flow with the upper reservoir, which also provided a natural gas-lift effect to support lower zone production.

This dual-reservoir smart completion approach allowed the well to deliver stabilized oil output, enhance zonal recovery, and reduce operational risk, while leveraging commingled production as a cost-effective alternative to installing additional artificial lift systems.

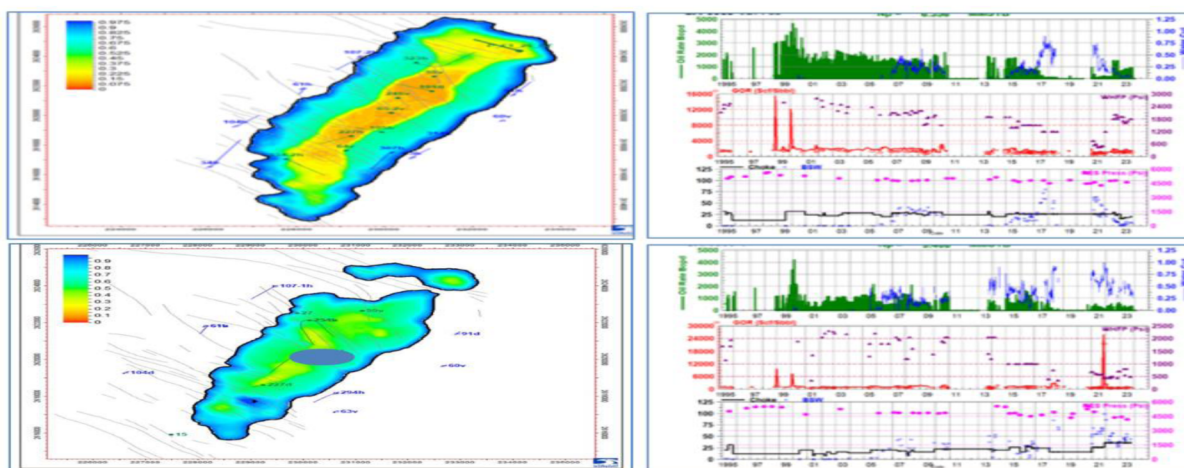


Figure 1—Historical Well Performance

## Smart Completion Design Methodology

The smart completion design was engineered to enable selective control of each reservoir interval while maintaining the flexibility for commingled production. The configuration included two production flow control valves (FCVs), each accommodating up to eight choke positions, allowing independent inflow management of the upper and lower zones. The lower assembly incorporated a wireline-retrievable plug and WEG nipple (2.25" F type) to provide contingency in the event of FCV malfunction, ensuring operational resilience.

## Zonal Isolation and Gauges

Production packers were installed across both the upper and lower reservoirs to provide mechanical isolation. Each zone was instrumented with solid gauge mandrels equipped with DHPT gauges (Tubing/Annulus) to continuously monitor downhole pressure and temperature. Two additional gauges were positioned above the top packer, ~100 ft TVD from the upper zone, to capture commingled flow behavior and provide reference data for real-time surveillance.

## Monitoring and Surveillance

The combination of multiple gauge stations across the completion enabled detailed zonal performance tracking. Data from the dual-sensor system supported accurate commingled flow calculations, feeding into the hybrid physics and data-driven workflow for zonal allocation and reservoir management.

## Contingency and Artificial Lift Provisions

A side pocket mandrel was placed below the upper FCV, offering a contingency production option in the event of FCV failure. In addition, four gas lift valves were installed to allow future artificial lift operations if required. This built-in flexibility ensured that the completion design could accommodate evolving reservoir conditions while minimizing the risk of downtime.

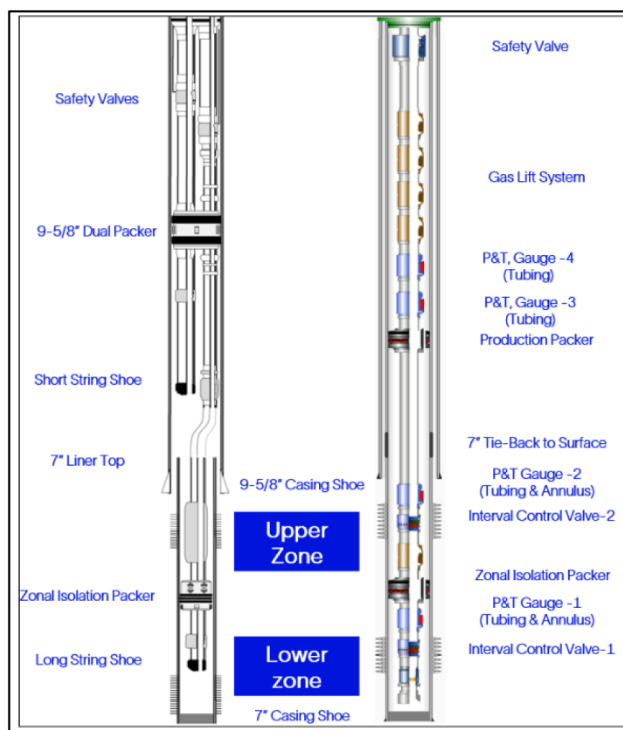


Figure 2—Pre and Post workover completion schematic

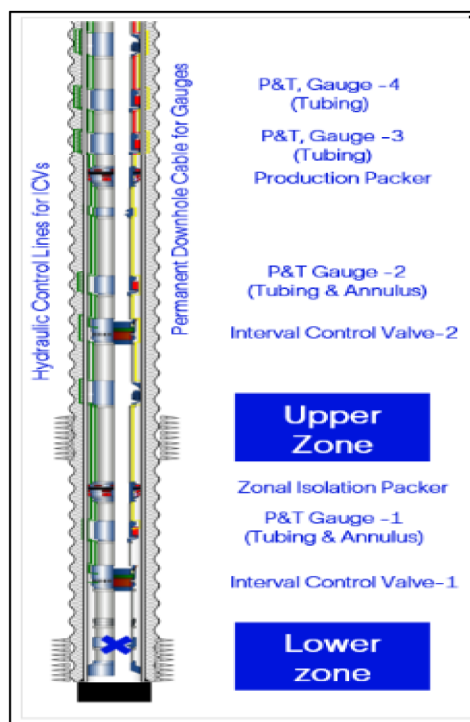


Figure 3—Smart completion key components

## Zonal Allocation Workflow

Accurate zonal flow rate estimation is critical for effective reservoir management and the evaluation of smart completions equipped with Inflow Control Valves (ICVs). To address this challenge, an in-house hybrid workflow was developed, combining physics-based modelling with data-driven analytics. This approach leverages real-time field data, including permanent downhole gauge (PDHG) measurements, ICV positions,

and surface production parameters, to dynamically allocate zonal production contributions and monitor well performance.

The workflow consists of three main steps (Figure 4).

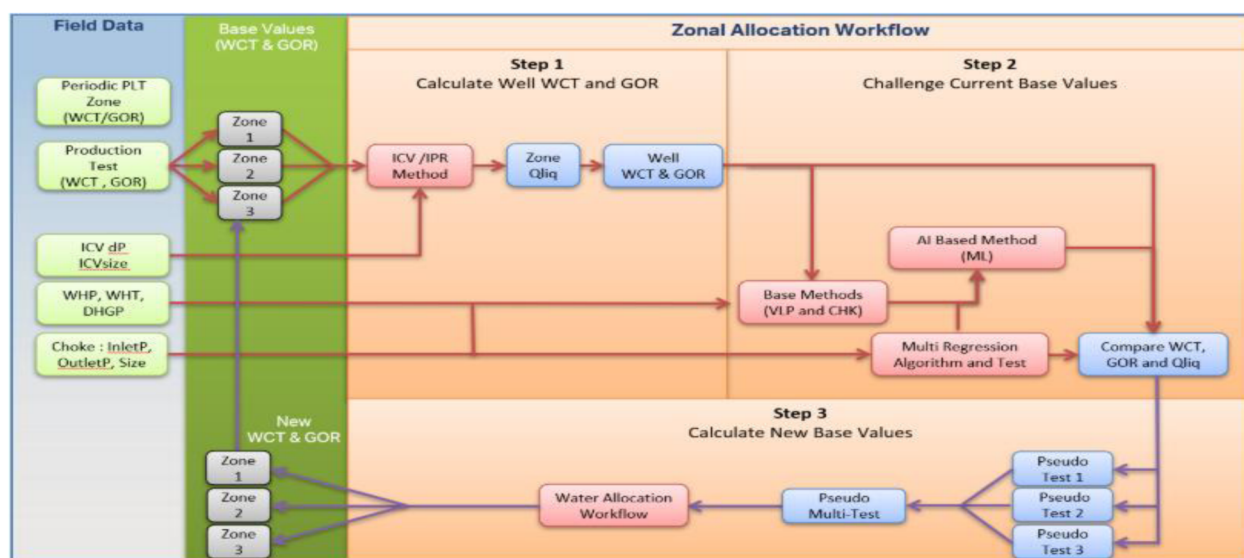


Figure 4—Zonal Allocation Workflow integrating physics-based and Data-driven methods

### Step 1: Calculate Well WCT and GOR

The objective of this step is to establish a baseline understanding of well performance using deterministic models.

- Each zone's inflow is modelled using its Inflow Performance Relationship (IPR).
- The pressure drop across the ICV is used to estimate zonal flow rates ( $Q_{iAP}$ ).
- Zone contributions are aggregated to calculate the overall Well Water Cut and Gas-Oil Ratio.

This step assumes steady-state conditions and relies on known reservoir and fluid properties to provide initial base values.

### Step 2: Challenge Current Base Values

The objective of this step is to validate and refine the base model using data-driven techniques.

- **Physics-Based Methods**
  - Vertical Lift Performance (VLP): Simulates pressure losses from bottomhole to surface.
  - Choke Model: Estimates flow through the surface choke using pressure differential and choke size.
- **Data Driven Method**
  - An analytical method embedded to ensemble learning based on historical test data and changes in ICV positions.
  - Inputs include ICV settings, downhole pressures, temperatures, and zonal flow rates.
  - Outputs include predicted WCT, GOR, and  $Q_i$  for each zone.

Physics-based and data driven outputs are compared against actual test data. Discrepancies are used to detect model drift or parameter inconsistencies.

### Step 3: Calculate New Base Values

The objective of this steps is to update the workflow with refined parameters for improved accuracy.

- Water Allocation Sub-Workflow uses the validated model to distribute water and oil production across zones.
- Pseudo-tests are performed to simulate different operating scenarios and recalibrate the model:
  - Pseudo Test 1: Upper zone open, lower zone closed.
  - Pseudo Test 2: Lower zone open, upper zone closed.
  - Pseudo Test 3: Both zones open with variable ICV positions.

These pseudo-tests enable dynamic recalibration, ensuring accurate zonal allocation under changing well conditions.

### Key Benefits

The integration of physics-based models with data-driven algorithms provides significant advantages for realtime reservoir surveillance and zonal production allocation. By continuously estimating the contributions of each interval in a commingled production environment, the workflow reduces reliance on intermittent production logging surveys and instead offers continuous allocation capabilities. This enables more reliable evaluation of ICV performance under variable operating conditions, including changes in valve settings or evolving reservoir behavior. The hybrid approach also enhances predictive capabilities by identifying potential phase changes and production anomalies before they impact well performance, allowing operators to make proactive decisions. Ultimately, the workflow improves the accuracy of reservoir management, enhances recovery efficiency, and supports operational optimization by providing actionable insights in a cost-effective manner.

## Smart Well Model Overview

The intelligent completion system is modelled using a physics-based software (Petroleum Experts GAP) that integrates multiple components of the well architecture to simulate and optimize zonal production. This model enables real-time surveillance, control, and dynamic allocation of flow from different reservoir zones through Intelligent Completion Valves (ICVs). The schematic ([Figure 5](#)) illustrates the smart well system, including zonal inflow, downhole gauges, choke configurations, and annulus/tubing interactions.

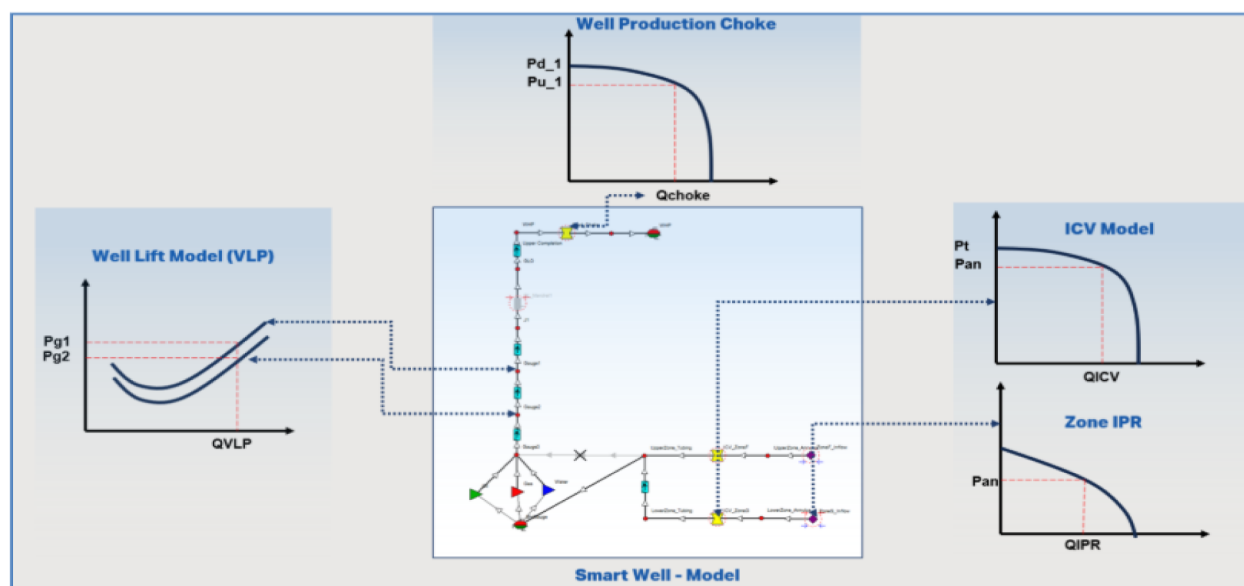


Figure 5—Smart Well Model Schematic

The smart well model incorporates the following subsystems:

- **ICVs for Zonal Control:** Adjustable valve settings regulate inflow from each zone, allowing selective production or commingled operation.
- **Downhole Gauges:** Permanent sensors monitor pressure and temperature in both tubing and annulus, providing continuous surveillance of zonal conditions.
- **Choke and Tubing Configurations:** The well production choke and tubing hydraulics are explicitly modelled to capture pressure drop behavior and optimize surface drawdown.
- **Annulus and Tubing Pressure Points:** Measurement points are integrated into the system to support commingled flow analysis and zonal allocation workflows.

This model serves as a backbone for simulating flow dynamics and optimise ICV valves settings. Sub-Models within the Framework

### 1. Well Lift Model (VLP)

- **Purpose:** Simulates the pressure-flow relationship along the tubing string.
- **Inputs:** Bottomhole pressures ( $P_{g1}$ ,  $P_{g2}$ ) and flow rate ( $Q_{VLP}$ ).
- **Output:** Predicts lift efficiency and determines optimal conditions for artificial lift or natural flow.

### 2. Well Production Choke Model

- **Purpose:** Evaluates choke performance and surface pressure drawdown.
- **Inputs:** Upstream and downstream pressures ( $P_{d\_1}$ ,  $P_{u\_1}$ ) and choke opening.
- **Output:** Flow rate ( $Q_{choke}$ ) through the choke, used for regulating wellhead production.

### 3. ICV Performance Model

- **Purpose:** Characterizes ICV response under varying flow and pressure regimes.
- **Inputs:** Total pressure ( $P_t$ ), annulus pressure ( $P_{an}$ ), and valve position.

- Output: Flow rate ( $Q_{ICV}$ ) through the valve, enabling selective zonal control.

#### 4. Zone Inflow Performance Relationship (IPR)

- Purpose: Models inflow from individual reservoir zones.
- Inputs: Annulus pressure ( $P_{an}$ ), reservoir properties, and ICV setting.
- Output: Zonal inflow rate ( $Q_{IPR}$ ), used to estimate each zone's contribution and manage reservoir performance.

This integrated smart well model forms the backbone of the surveillance and optimization strategy. By combining tubing hydraulics, surface choke dynamics, ICV performance, and zonal inflow relationships, the model provides:

- Accurate simulation of well dynamics under different ICV settings.
- Improved capability to optimize valve positions for selective or commingled production.
- Enhanced reservoir management by predicting phase behavior and production anomalies.
- A robust foundation for linking field data with the hybrid physics and AI-driven zonal allocation workflow.

### Reservoir Monitoring Plan (RMP)

A comprehensive Reservoir Monitoring Plan (RMP) was designed and executed to acquire the necessary data for validating the zonal production allocation workflow. The monitoring strategy combined production testing at various Intelligent Completion Valve (ICV) positions with Production Logging Tool (PLT) measurements to establish a robust dataset against which the hybrid physics- and data-driven allocation model could be calibrated and validated.

**Surface Production Testing (PTS).** Controlled production tests were carried out to characterize reservoir contributions under varying ICV choke positions. The upper reservoir was tested independently at all choke settings, providing a benchmark inflow profile, whereas the lower reservoir could not be produced in standalone mode due to its high water cut (~70%). To address this constraint, commingled production tests were conducted with both reservoirs open, enabling evaluation of zonal interactions and overall system performance.

**Production Logging (PLT).** Complementing the surface tests, production logging surveys were performed under commingled flow conditions to quantify the split between upper and lower zone contributions. These measurements provided direct validation of the hybrid physics- and data-driven allocation model, confirming good agreement between predicted and measured zonal inflows. The PLT data also demonstrated the effectiveness of ICV manipulation in controlling inflow distribution, delaying unwanted water production, and enhancing the efficiency of commingled operations.

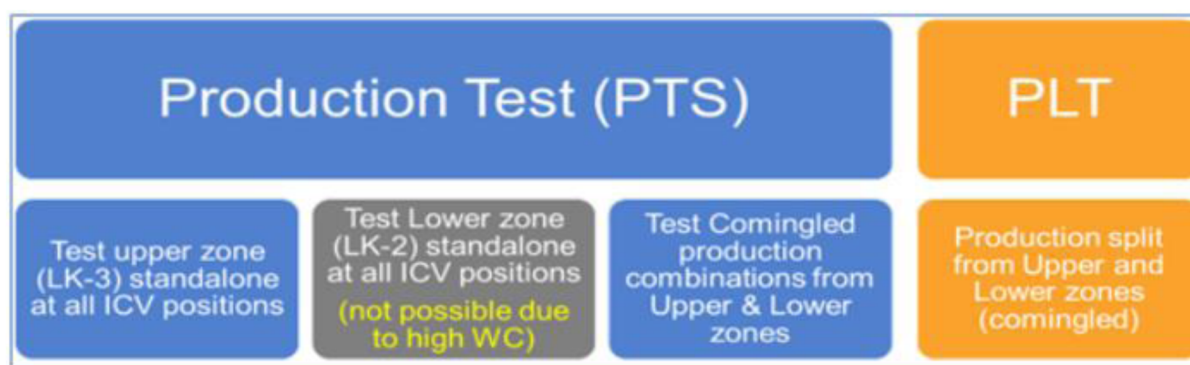


Figure 6—Executed Commissioning Plan showing the integrated production testing (PTS) and PLT validation program

## Validation and Outcomes

The combined PTS and PLT dataset allowed rigorous validation of the zonal allocation workflow. Comparison between measured and predicted zonal rates demonstrated high accuracy in the model's ability to allocate production contributions under different ICV settings and well conditions. This validation not only confirmed the technical robustness of the allocation methodology but also reinforced confidence in using the workflow for continuous optimization without the need for frequent logging interventions.

Economically, the implementation of the RMP and smart completion strategy resulted in an additional ~300 BOPD of incremental production and avoided approximately \$4.8MM in CAPEX through reduced workover requirements and improved reservoir management efficiency.

### Upper Zone Production Testing and Validation

The upper reservoir was subjected to a sequence of controlled production tests at different ICV choke settings to characterize its inflow performance and establish a reference profile. The test sequence demonstrated clear variations in production rates as the ICV was manipulated across multiple positions, ranging from more restricted settings to fully open conditions.

Oil production rates were observed between 950 STB/d and 1860 STB/d, depending on the choke position. At more restrictive ICV openings, rates stabilized around 950 -1100 STB/d, while progressively wider choke settings increased output to between 1500 – 1860 STB/d. Intermediate ICV positions delivered incremental step changes in flow, with values typically in the 1300 -1730 STB/d range. These results confirm that the upper zone provides a strong and stable contribution to commingled production, and that ICV manipulation allows fine-tuning of zonal inflow to balance performance against reservoir management objectives.

The surface pressure response was also consistent with the expected hydraulic behavior of the ICV system, validating both the mechanical integrity of the completion and the calibration of the smart well model. The obtained inflow profile from these tests served as a benchmark for subsequent zonal allocation workflows and provided reference data for validating commingled production scenarios when both reservoirs were open.



Figure 7—Upper Zone Production Testing showing oil rate variations across different ICV positions

The developed allocation workflow was tested against real-time data acquired during the production testing of the upper reservoir. The objective was to validate the accuracy of the hybrid physics- and data-driven model by comparing predicted zonal production rates with measured data from the surface production tests (PTS).

Figure 8 illustrates the results of this comparison. The model-calculated production rates closely tracked the measured liquid rates across the full range of ICV choke positions, demonstrating a strong correlation between predicted and observed performance. The allocation workflow successfully captured both the step changes in flow associated with valve adjustments and the stabilized production plateaus at each choke setting.



Figure 8—Upper Zone Allocation Results showing model-calculated versus measured production rates

This good agreement confirms the robustness of the model in simulating zonal inflow dynamics under varying operating conditions. The validation also highlights the ability of the workflow to reduce reliance on frequent production logging by providing continuous, high-fidelity allocation results directly from real-time surveillance data. Ultimately, the results establish confidence in using the workflow for ongoing optimization of ICV settings and reservoir management decisions.

## Commingled Production Testing and Validation

Following the standalone evaluation of the upper reservoir, commingled production tests were performed with both the upper and lower zones flowing simultaneously. The purpose of these tests was to characterize overall well performance under dual-reservoir conditions and to provide calibration data for validating the zonal allocation workflow.

Figure 9 presents the measured liquid rates during the commingled testing sequence. Oil production stabilized between 1560 STB/d and 2330 STB/d, depending on the ICV choke positions and flow configurations. As expected, the overall liquid rate increased compared with the upper-zone-only production tests, reflecting the contribution of the lower reservoir despite its high water cut (~70%). The dynamic pressure response captured during the test also highlighted the zonal interactions within the wellbore and confirmed that the upper reservoir acted as a natural gas-lift source for the lower zone.



Figure 9—Commingled Production Testing showing stabilized liquid production rates across varying ICV positions

The commingled production tests provided critical input for the hybrid allocation model. By comparing predicted zonal contributions with actual measured rates, the workflow could be validated under realistic multizone operating conditions. These results confirmed that the smart completion system, when operated in commingled mode, successfully balanced inflows from both zones, optimized total production, and managed the water influx from the lower reservoir.

The hybrid allocation workflow was further validated using real-time surveillance data and production test results acquired during commingled production from the upper and lower reservoirs. The objective was to assess the model's ability to accurately predict both total liquid rate and water cut under dual zone flowing conditions.

As shown in Figure 10, the workflow predictions exhibited a strong match with the measured field data. The calculated total liquid production closely tracked the observed rates across multiple ICV choke positions and operational scenarios, successfully replicating both transient changes and stabilized plateaus. Similarly, the predicted water cut showed good alignment with the measured values, confirming that the model effectively captured the contribution of the high-water-cut lower reservoir within the commingled stream.

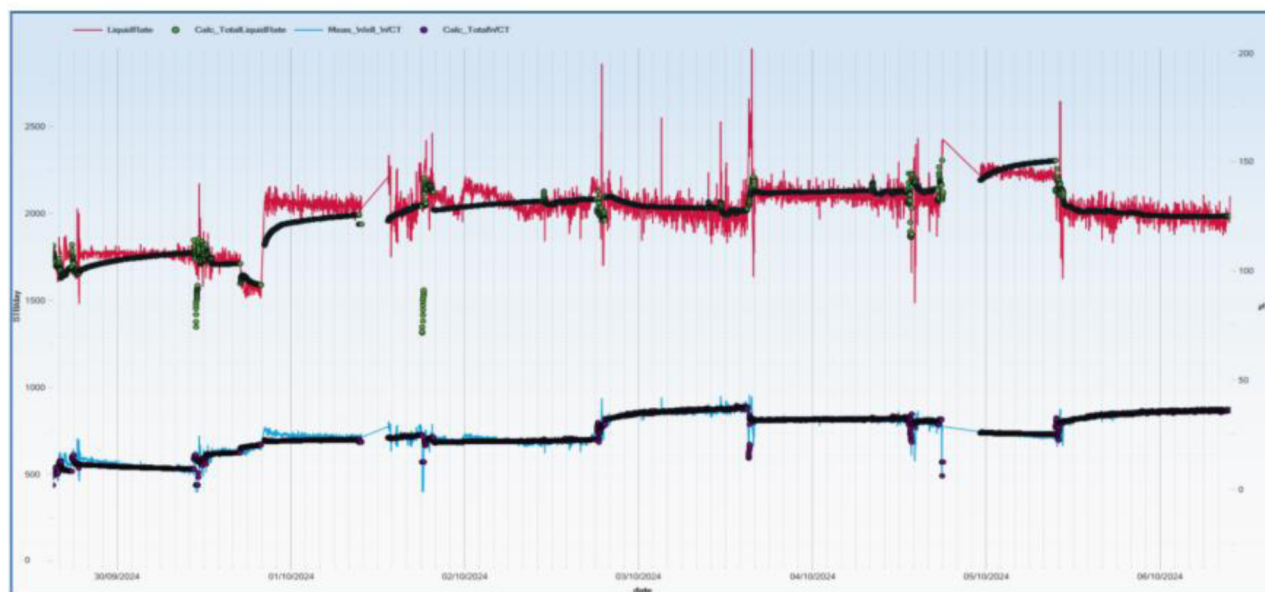


Figure 10—Commingled Allocation Results showing model-calculated versus measured liquid rate and water cut.

This strong agreement between model and measured data demonstrates the robustness of the workflow in dynamically allocating zonal contributions under complex operating conditions. By integrating real-time field measurements with the physics- and data-driven modelling framework, the allocation method provided continuous and reliable surveillance without the need for frequent logging interventions. The validated performance of the workflow under commingled operation highlights its potential as a practical reservoir management tool, enabling proactive optimization of ICV settings and improved control of water production.

### Production Profiles (PLT) and Workflow Validation

To further validate the allocation workflow, a Production Logging campaign was performed while the well was producing under commingled conditions and at different ICV positions. The PLT measurements provided direct insight into zonal flow profiles, enabling comparison between observed contributions and workflow-predicted results.

The PLT surveys confirmed that both the upper and lower reservoirs were contributing to commingled production, with the upper zone consistently showing a higher liquid fraction compared to the lower zone.

Across the tested ICV positions, the upper zone's liquid contribution ranged from approximately 40-67%, while the lower zone contributed between 33-60%, depending on the choke configuration and operating conditions. These variations reflect the expected dynamics given the lower zone's high water cut (~60-65%).

The workflow's predictions of zonal liquid contributions were compared against the PLT results for each ICV position (Figure 12). The agreement was strong, with model estimates closely tracking the measured split from the logging data. Both the overall liquid rate and the relative zonal fractions showed minimal deviation, confirming that the hybrid physics- and data-driven approach is capable of accurately replicating field-measured behavior under commingled operation.

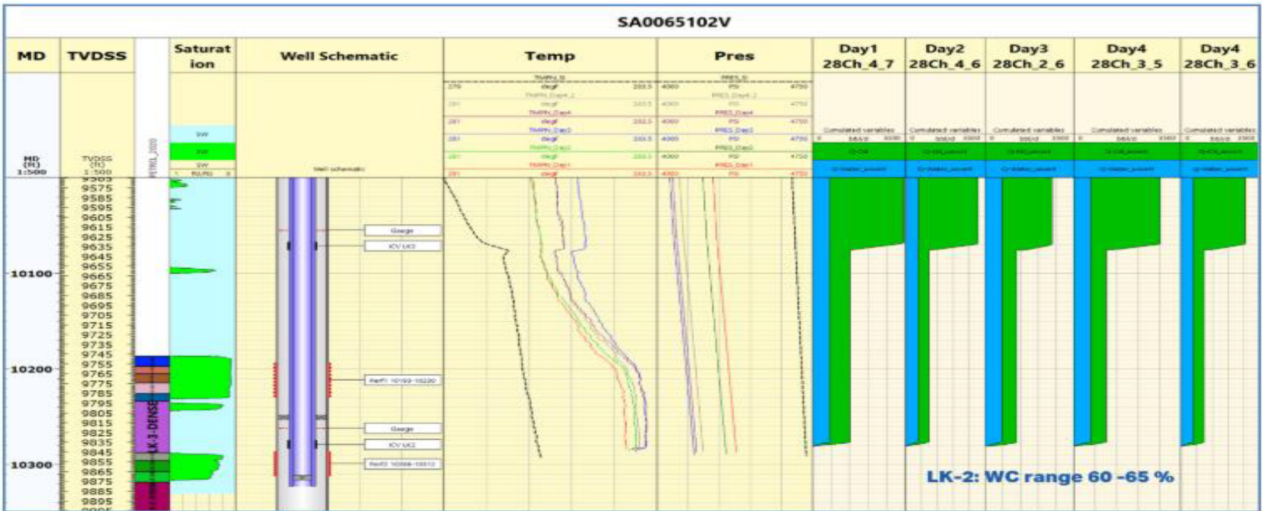


Figure 11—Production Logging results showing zonal contributions from the upper and lower reservoirs.

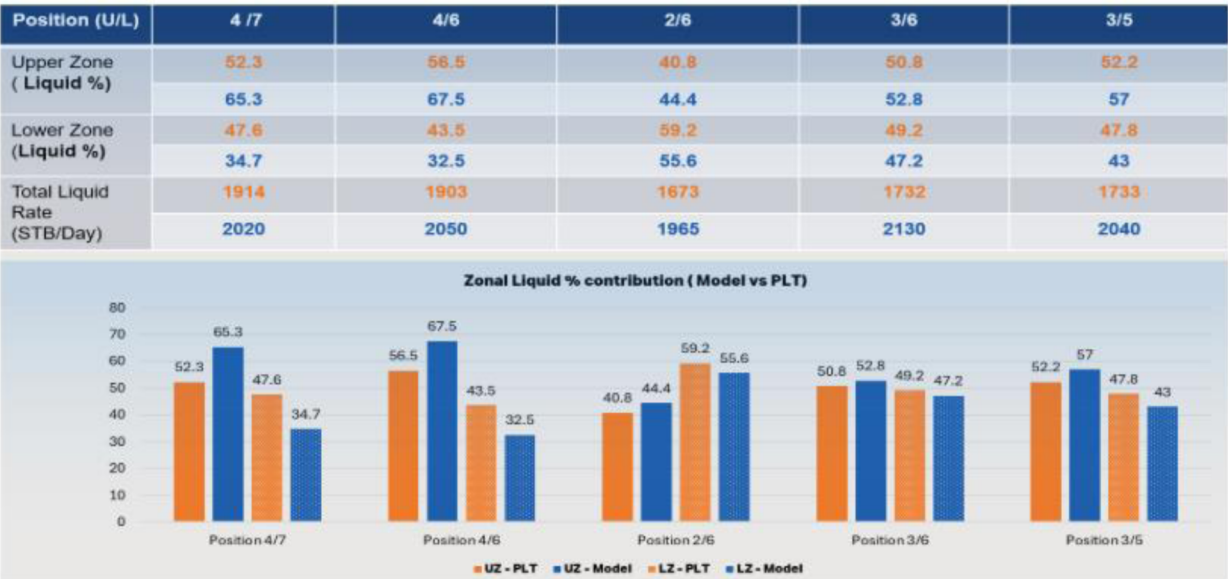


Figure 12—Zonal Allocation Results comparing model predictions against PLT measurements.

Validation Outcomes

This validation exercise reinforced confidence in the allocation workflow as a reliable surveillance tool. By aligning well with PLT measurements, the workflow demonstrated its potential to reduce reliance on frequent logging surveys while still providing continuous, real-time zonal allocation. The integration of PLT-confirmed performance with model-based surveillance ensures that ICV manipulation strategies can be optimized with greater accuracy, ultimately improving reservoir management and water control.

Smart Completion Solution Validation and Best Practices

The surveillance and allocation workflow implemented in this pilot has been successfully validated and demonstrated as a reliable tool for real-time reservoir management. As part of the company's broader digitalization strategy, the workflow is being integrated into a digital platform to enable continuous monitoring and automated optimization of smart well performance. This integration ensures that real-time data streams are consistently leveraged for production allocation, ICV control, and proactive decision-making.

To maintain long-term sustainability and accuracy, several best practices have been identified and are recommended for future deployment of similar workflows:

- **Continuous Surveillance:** Pressure and temperature trends should be monitored in real time, supported by automated event detection and alarm notifications to flag deviations from expected behavior.
- **Surface Measurements:** Periodic surface separator tests are necessary when significant changes in well conditions occur, providing reference points to recalibrate model predictions.
- **Model Recalibration:** Both choke and hydrostatic models should be recalibrated when operational conditions shift to maintain consistency between predicted and measured values.
- **Cross-Validation:** Allocation results should be cross-checked against PI-based reservoir models to strengthen confidence in predictions.
- **Gradient Consistency Checks:** Regular validation of pressure and fluid gradients ensures model integrity and highlights potential anomalies in well performance.

Adhering to these practices ensures that the hybrid physics- and data-driven workflow remains aligned with field realities, enhancing predictive reliability and operational efficiency. By embedding such practices into the reservoir surveillance strategy, operators can maximize the value of smart completions, reduce the need for frequent logging interventions, and sustain accurate zonal allocation over the life of the well.

## Key Outcomes and Impact

The pilot testing confirmed the effectiveness of the smart completion and allocation workflow in managing and understanding zonal production within a commingled well, without the need for frequent interventions. The results demonstrated how petroleum and reservoir engineers can leverage automated data processing and hybrid workflows to transform large volumes of surveillance data into actionable insights, ultimately optimizing production and maximizing reservoir performance.

More importantly, the integration of the digital allocation system with the inflow control hardware enabled a level of precision, responsiveness, and real-time insight that traditional methods could not deliver. By combining continuous surveillance, predictive modelling, and automated validation, the digital workflow fundamentally changed how zonal performance was interpreted, controlled, and optimized.

From a business impact perspective, the case study delivered measurable results, including incremental oil gains of ~300 BOPD and significant CAPEX savings by reducing the need for major workovers or new well replacements. The reservoir monitoring plan and real-time allocation workflow also reduced dependency on production logging, further lowering operational costs.

This project highlights the growing importance of digital technologies in modern reservoir management. As wells become more complex and increasingly data-rich, the integration of smart completion hardware with advanced digital workflows will be essential to unlock their full potential, ensuring both improved recovery efficiency and sustainable long-term field development.

## Conclusions

This case study successfully demonstrated the deployment of smart selective completion technology to enable commingled production from dual reservoirs. The integration of intelligent hardware with a hybrid surveillance and allocation workflow allowed real-time optimization of zonal inflows, improved reservoir management, and measurable production gains. The implementation resulted in an incremental ~300 BOPD oil gain and avoided approximately \$4.8MM in CAPEX through reduced workover and intervention requirements. To fully capitalize on the smart completion capabilities, the use of a permanent control panel is recommended for future deployments.

The results confirm the technical feasibility and operational value of applying smart completions in complex well environments. The project highlighted the importance of integrated planning, rigorous simulation, and adaptive execution to restore production and extend well life. Lessons learned from this pilot provide a practical framework that can guide future smart well implementations in similar reservoir settings.

The novelty of this work lies in combining smart completion technology with a digital, real-time surveillance and zonal allocation workflow, demonstrating clear benefits in precision, responsiveness, and efficiency compared to conventional approaches. By bridging physics-based models with data-driven validation, the workflow transformed the way zonal performance was monitored and optimized.

Overall, this paper provides valuable insights and best practices that can support wider adoption of smart completions across the industry. The outcomes highlight how digital technologies, when coupled with advanced completion systems, can unlock additional hydrocarbon recovery and deliver sustainable reservoir management in increasingly complex and data-rich operating environments.

## **Acknowledgment**

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